

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		Lines up to satellite and down to C Ignore arrows and any additional paths shown		1		1		
	(b)		Distance = $4 \times 36\,000$ [km] (1) = 144 000 k[m] (1) = 144 000 000 [m] (1) N.B. Conversion mark can be awarded on incorrect calculations even 36 000 000 Answer of 1.44×10^n where n is not equal to 8 award 2 marks		3		3	3	
	(c)		Speed = $\frac{\text{distance}}{\text{time}} = \frac{144\,000\,000 \text{ecf}}{0.48}$ or $\frac{36\,000\,000}{0.12}$ or $\frac{72\,000\,000}{0.24}$ (1) = 300 000 000 [m/s] (1)	1	1		2	2	
			Question 5 total	1	5	0	6	5	0